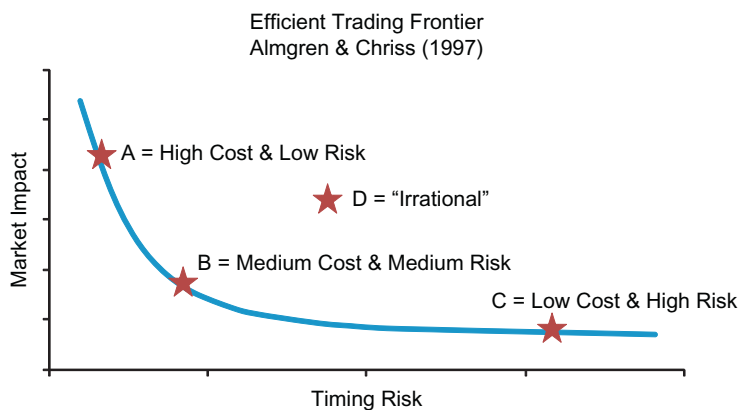


better prices. For example, an algorithm may call for a 15% POV rate in the morning, increasing to a POV of 25% beginning midday (back-loading) to take advantage of expected better prices in the afternoon while still balancing the trade-off between market impact and timing risk. Furthermore, an algorithm may call for a POV of 30% in the morning, falling to 10% in the afternoon (front-loading) as a means to reduce risk and hedge the trade list.

Solving the optimization problem described above for various levels of risk will result in numerous optimal trading strategies. Each has the lowest cost for the specified level of risk and the lowest risk for the specific amount of cost. The set of all these optimal strategies comprises the efficient trading frontier (ETF) first introduced by Almgren and Chriss (1999).

After computing the ETF, investors will determine the most appropriate “optimal” strategy for their implementation goal. For example, informed traders with expectations regarding future price movement are likely to select an aggressive strategy (e.g., $\text{POV} = 30\%$) with higher cost but more certainty surrounding expected transaction prices. Indexers are likely to select a passive strategy (e.g., $\text{POV} = 5\%$) or a risk neutral strategy to reduce cost. Some investors may select a strategy that balances the trade-off between cost and risk depending upon their level of risk aversion, and others may elect to participate with volume throughout the day (e.g., VWAP strategy).

It is important that investors thoroughly evaluate alternative strategies to determine the one that provides the highest likelihood of achieving their investment goal. Figure 1.13 shows four different “optimal” strategies.



■ **Figure 1.13** Efficient Trading Frontier.
Almgren and Chriss (1997)